

CMPE 272 Midterm Study Guide

Section 49, Fall 2026

The exam

- Wednesday, October 7, in class, after the project abstract presentations.
- It covers sessions 1 to 7.
- It is a Canvas quiz: "Mid-term Exam- Requires Respondus LockDown Browser". Closed book. 90 minutes. One attempt. 10% of the course grade.
- The access code is given in class when the exam opens.
- Install Respondus LockDown Browser before class and take "Sample Quiz- Requires Respondus LockDown Browser" on Canvas to check that it works on your laptop.

How to use this guide

Each session below lists what you should be able to explain, with the deck and slide to go back to. The decks are in Canvas Files, under Lectures. Slide numbers are the page numbers of the PDFs posted to section 49. Work through the self-check questions without your notes, then check yourself against the slides.

To test yourself, take "Sample Midterm (FA26 practice)" under Quizzes. It counts for nothing, you can take it as often as you like, and it does not need LockDown Browser. It shows the right answer and the slide after you submit. It closes at 6:00 PM on October 7. Its questions are practice questions, not the exam's.

Session 1: Introduction, enterprise architecture, Ansible

Be able to explain:

- What makes software an enterprise application, and why enterprise architecture is needed. (Lecture-01, slides 20 and 21)
- The two enterprise architecture frameworks named in class, Zachman and TOGAF, and what TOGAF consists of. (Lecture-01, slides 22 and 70)
- What Ansible automates, how it reaches hosts, and why it needs no agent. (Lecture-01, slides 43 and 44)
- Ansible's components: inventory, modules, plugins, playbooks. (Lecture-01, slide 45)
- Ad-hoc commands against playbooks; playbook sections, variables, facts, privilege escalation, tasks and handlers, roles. (Lecture-01, slides 53 to 65)

Self-check:

1. What does an enterprise architecture framework divide an architecture description into?
2. What is a handler in a playbook, and when does it run?
3. Why are roles useful when several playbooks need the same setup?

Session 2: Enterprise infrastructure

Operating systems, identity and storage (deck Lecture-02a):

- Types of operating systems, and the OS services an enterprise runs. (Lecture-02a, slides 16 and 48)
- What systemd is, and what it replaces. (Lecture-02a, slide 36)
- Load averages and the tools that show them. (Lecture-02a, slide 49)

- Identity: SAML, OAuth and OpenID Connect, identity providers, Active Directory roles. (Lecture-02a, slides 55 to 59; 66)
- File systems: the virtual file system, the inode, why file systems prefer RAM and sequential I/O, the page cache. (Lecture-02a, slides 81 to 89)
- Disks and RAID. (Lecture-02a, slides 98 and 106)
- Block, file and object storage: how each is accessed, what each is good for. (Lecture-02a, slides 109 to 112)
- Puppet, Chef and Ansible; declarative against imperative configuration. (Lecture-02a, slide 124)

Networking (deck Lecture-02b):

- The OSI model and IP addressing you are expected to know already. (Lecture-02b, slide 5)
- QUIC: what it adds on top of its transport, and how it avoids head-of-line blocking. (Lecture-02b, slides 7 and 8)
- Switching, VLANs, trunks and 802.1Q tagging, including the native VLAN. (Lecture-02b, slides 39 to 45)
- How a switch builds its forwarding (CAM) table, and what it does with an unknown destination. (Lecture-02b, slide 53)
- Spanning Tree. (Lecture-02b, slide 54)
- Routing: IGP against EGP, BGP, eBGP and iBGP, multi-homing to one or more providers. (Lecture-02b, slides 65, 79 and 88)
- Data center billing: committed information rate against 95th-percentile metering. (Lecture-02b, slides 104 and 105)
- Control plane against data plane. (Lecture-02b, slide 109)
- Data center designs: Clos networks, the traditional design, leaf/spine, Cisco ACI. (Lecture-02b, slides 110 to 114)
- NAT: inside and outside addresses; static NAT, dynamic NAT and overloading (PAT). (Lecture-02b, slide 136)

Self-check:

1. Compare block, file and object storage: the access protocol and one typical use for each.
2. When are dirty pages in the page cache written to disk?
3. Which of SAML, OAuth and OpenID Connect delegates access, and which ones carry authentication?
4. Which plane builds the routing table, and which one forwards packets?
5. Which billing scheme lets a customer burst above the committed rate?
6. How can many inside hosts share one public address?

Session 3: Application frameworks and APIs

- Legacy and modern inter-process communication: RPC, CORBA, SOAP, WSDL, REST, Thrift. (Lecture-03, slide 6)
- REST: the properties it aims for and its architectural constraints. (Lecture-03, slides 7 to 9)
- Protocol Buffers and gRPC: serialization, transport, the four call types. (Lecture-03, slides 10 and 11)
- The enterprise service bus, message buses and message queues. (Lecture-03, slides 12 to 14)
- Enterprise Integration Patterns: channels, message construction, routing, transformation, endpoints, and the Rider Auto Parts example. (Lecture-03, slides 17 and 18; 28 to 37)
- Java EE application servers and the Spring framework. (Lecture-03, slides 24 and 25; 39)
- Inversion of control and dependency injection. (Lecture-03, slides 41 and 42)

- Event-driven architecture: events, brokers. (Lecture-03, slides 43 and 44)
- Kafka, Pulsar, Dapr and Temporal. (Lecture-03, slides 46 to 50)
- DevOps and the "wall of confusion"; what makes an architecture microservices. (Lecture-03, slides 53 to 59)
- Service mesh: what it is, the pattern, per-node against sidecar, the three generations, AWS App Mesh. (Lecture-03, slides 64 to 73; 80)
- Contract-first API design, OpenAPI 3.1, versioning, design to mock to generate to test. (Lecture-03, slides 92 to 96)

Self-check:

1. Name gRPC's four call types.
2. How are inversion of control and dependency injection related?
3. Give one drawback of the per-node proxy model.
4. What is the difference between a discrete event and a continuous event?
5. Why design the API contract before writing the code?

Session 4: Cloud-native services

- What microservices are, their key characteristics, their benefits and their costs. (Lecture-04, slides 6 to 12)
- SLI, SLO, SLA and the error budget. (Lecture-04, slide 13)
- Synchronous against asynchronous communication, the monolith style, and the orchestrated and choreographed variants with their trade-offs. (Lecture-04, slides 14 to 25; 26)
- Integration concerns: contract, delivery, reliability, state, observability. (Lecture-04, slides 34 to 38)
- Safety measures: the circuit breaker and idempotency. (Lecture-04, slides 40 and 41)
- Scaling: load balancer types, scaling databases, caching, service discovery. (Lecture-04, slides 42 to 48)
- Containers, orchestration and the service mesh in a cloud-native platform. (Lecture-04, slides 49 to 51)
- Conway's Law, the Inverse Conway's Law, technical sprawl. (Lecture-04, slides 55 and 56)
- Serverless: what it means, what triggers code, where the savings come from, and the Prime Video case. (Lecture-04, slides 57 to 62; 66)

Self-check:

1. Define SLI, SLO, SLA and error budget. How much downtime per year does "five nines" allow?
2. Name two costs of microservices.
3. Give one trade-off of synchronous calls and one of asynchronous events.
4. What does the Inverse Conway's Law ask of the organization?
5. What did the Prime Video case show?

Session 5: Platform engineering and the delivery lifecycle

- Internal developer platforms and the golden path. (Lecture-05, slide 6)
- The four DORA metrics, and how to use them. (Lecture-05, slide 7)
- Governance and compliance as code. (Lecture-05, slides 8 and 25)
- The SDLC: what it is, its stages and milestones, the popular models. (Lecture-05, slides 11 to 17)
- DevOps: principles and technologies. (Lecture-05, slides 22 and 23)

- Scrum and Kanban: roles, events, boards, Kanban values. (Lecture-05, slides 26 to 34)
- Jenkins: pipelines, pipeline as code, plugins, build triggers, static analysis, test reports, SonarQube. (Lecture-05, slides 36 to 55)
- Continuous integration, continuous delivery, branching, environments, the deployment pipeline. (Lecture-05, slides 58 to 72)
- GitHub Projects. (Lecture-05, slide 76)

Self-check:

1. List the four DORA metrics. Why are they not meant for individual performance reviews?
2. What is the difference between continuous integration and continuous delivery?
3. What does "pipeline as code" mean in Jenkins, and where is the pipeline definition kept?
4. What is a golden path?

Session 6: Enterprise applications: ERP, SCM and CRM

- SAP, Salesforce and Dynamics today; extending without forking; why replacing an ERP is hard. (Lecture-06, slides 5 to 7)
- Functional structure and the silo effect. (Lecture-06, slides 9 and 10)
- Client-server, service-oriented and ERP architectures. (Lecture-06, slides 15 to 17)
- What an ERP is, how it works, its components and modules; before and after ERP. (Lecture-06, slides 19 to 23; 25 to 28)
- Pros and cons of enterprise systems and ERP packages. (Lecture-06, slides 31 to 33)
- Master data against transactional data, and creating an order in SAP. (Lecture-06, slides 39 to 45)
- CRM: what it does, CRM against SFA, benefits. (Lecture-06, slides 46 to 58)
- Cloud against on-premises ERP, hybrid solutions, how to implement an ERP. (Lecture-06, slides 60 to 67; 68)

Self-check:

1. Give one example each of master data and transactional data from a sales order.
2. What is the silo effect, and how does an ERP address it?
3. Why is a strangler-fig replacement safer than a big-bang cutover?
4. How does CRM differ from SFA?

Session 7: Business processes, messaging and service composition

- UML: the diagram types used, multiplicity, association, aggregation and composition; class, use case, activity and sequence diagrams. (Lecture-07, slides 9 to 12; 13 to 16)
- ER diagrams: notation, cardinality and participation, primary and foreign keys. (Lecture-07, slides 21 to 24)
- BPMN: business process modeling, symbols, pools and lanes, activities, events, gateways, the pizza example. (Lecture-07, slides 26 to 40; 41)
- BPM platforms and Camunda 8. (Lecture-07, slides 44 to 46)
- Messaging: why not call the next service, queue or topic, delivery guarantees, idempotency keys, ordering by key, retries and the dead-letter queue. (Lecture-07, slides 49 to 54)
- The dual-write problem and the outbox pattern; Kafka; SQS, SNS and EventBridge; schemas; commands, events and change data capture; the idempotent consumer. (Lecture-07, slides 55 to 60)
- Composition: orchestration or choreography, sagas, workflow engines, where a BPMN engine

fits, pitfalls. (Lecture-07, slides 62 to 66)

Self-check:

1. What is the difference between cardinality and participation?
2. Compare exclusive, inclusive and parallel gateways.
3. A queue and a topic each receive one message. Who gets it in each case?
4. Why must a consumer acknowledge only after its side effect is committed?
5. Explain the outbox pattern in two sentences.
6. What is a compensating action in a saga?